DS4420: Machine Learning and Data Mining 2
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# New MovMie: Leveraging Collaborative Filtering and a Multilayer Perceptron Network for Movie Recommendations

[redacted]

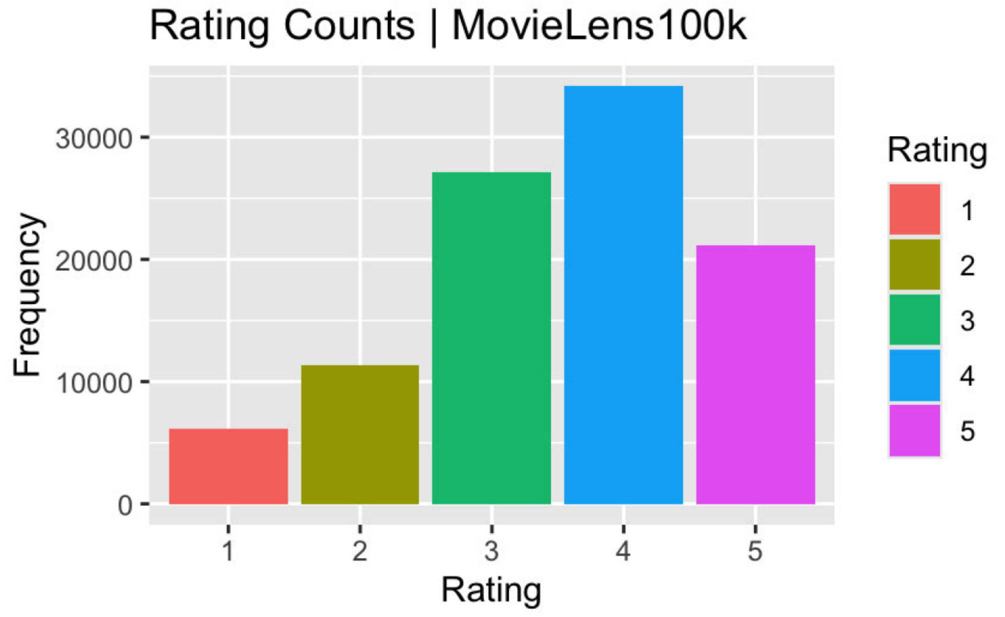
## Purpose

- To evaluate collaborative filtering (CF) and multilayer perceptron (MLP) applications to movie recommendations.
- To combine CF and MLP to identify a movie recommendation for the user that they would not try on their own but would likely enjoy.

## Background

- Addictive algorithms on video streaming platforms and the like cause echo chambers of information, doom-scrolling or watching, and impact mental health and work efficiency.
- Previous work explores CF and MLP for movie recommendation systems, but this combination is unprecedented in this manner.

Rating Counts | MovieLens100k




### Data Overview

- MovieLens 100k dataset
- Rating distribution:
  - {1: 6110, 2:11370, 3:27145, 4:34174, 5:21201}
- Released 4/1998
- CF Data:
  - 1000 users, 1700 movies
- MLP Data:
  - Rating, timestamp, genre, title, age, gender, occupation, zipcode, bias

## Methods

**User-User Collaborative Filtering (UUCF)**
UUCF makes recommendations based on similar users under the assumption that users who liked similar movies in the past will continue to do so in the future.

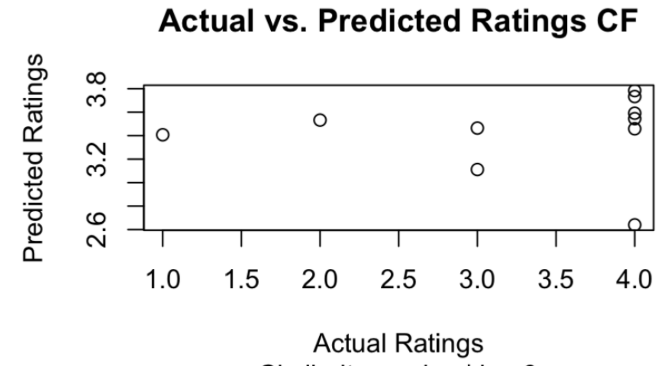
**Multilayer Perceptron (MLP)**
MLP takes input data and passes it through layers, which learn patterns in data by adjusting weights through backpropagation.

**UUCF + MLP Hybrid Model**
- Generated datasets of [user | movie | rating] from collaborative filtering and MLP
- Extracted movies the user has not rated, the MLP scores, and the CF scores
- Evaluated scores, returning movie with the highest CF rating

## Results

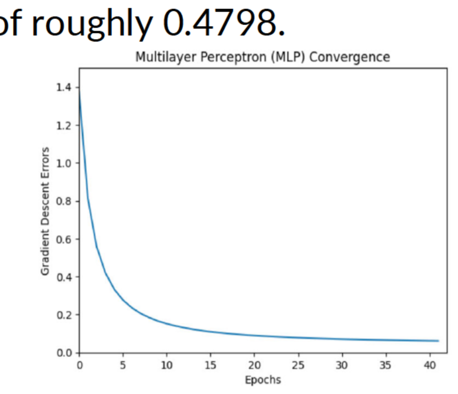
**User-User Collaborative Filtering (UUCF)**
The best performing hyperparameters were *cosine* for the similarity metric and *3* for the k-value determined through RMSE (1.05), MAE (0.77), and rounded accuracy (0.6) metrics on a test set of 9 masked users.

Actual vs. Predicted Ratings CF




Similarity: cosine | k = 3

**Multilayer Perceptron (MLP)**
Utilizing a step size of 0.1, an input layer with 2500 nodes, a single hidden layer with 10 nodes, log loss objective function, and the ReLU activation function, and an output with a single node (movie rating) and the sigmoid activation function, the algorithm produced an accuracy of roughly 0.4798.




**UUCF + MLP Hybrid Model**
- Preferred and recommended genres differed
- CF score consistently higher than MLP score, with MLP scores low
- For users with preferred and recommended genres with overlap, CF score was still higher than MLP, indicating success

Comparing the top 85 movies for User: 244


| User | Recommended Movie | CF Score | MLP Score | Preferred Genre(s) | Rec. Movie Genre |
|---|---|---|---|---|---|
| 196 | ID: 357 Title: One Flew Over the Cuckoo's Nest (1975) | 4.246564 | 4.0 | Comedy | Drama |
| 186 | ID: 286 Title: The English Patient (1996) | 4.609684 | 4.0 | Thriller | Drama, Romance, War |
| 22 | ID: 183 Title: Alien (1979) | 4.334328 | 4.0 | Action, Drama | Action, Horror, Sci-Fi, Thriller |
| 244 | ID: 483 Title: Casablanca (1942) | 4.67168 | 4.0 | Comedy | Drama, Romance, War |
| 166 | ID: 228 Title: Star Trek: The Wrath of Khan (1982) | 3.899833 | 3.0 | Thriller, Action, Mystery | Action, Adventure, Sci-Fi |
| 298 | ID: 192 Title: Raging Bull (1980) | 4.451612 | 4.0 | Feel-good movies (i.e. Children's, Comedy, Romance) | Drama |
| 115 | ID: 169 Title: The Wrong Trousers (1993) | 5.0 | 4.0 | Drama, Action | Animation, Comedy |
| 253 | ID: 195 Title: The Terminator (1984) | 4.127183 | 4.0 | Attention-grabbing movies (i.e. Action, Adventure) | Action, Sci-Fi, Thriller |
| 305 | ID: 23 Title: Taxi Driver (1976) | 4.355715 | 4.0 | Drama | Drama, Thriller |
| 6 | ID: 654 Title: Chinatown (1974) | 4.671806 | 4.0 | Drama | Film-Noir, Mystery, Thriller |

## Conclusions

- There are trends in the hybrid model's outputs (comedy/feel-good → drama and vice versa).
- Hybrid model CF + MLP offers promising results for suggesting a movie the user would not try on their own assuming they think similarly to MLP
- Hybrid model also indicates potential for the movie to be highly rated due to the recommendation being made on account of similar users' enjoyment

## Next Steps

- Test hybrid model on additional data, subsets of this data
- Test hybrid model on newer data with additional characteristics
- Adjust algorithm to apply to short-form videos (such as TikTok, Instagram Reels, etc.)

## References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